

Chorus
GPUs for ANSI Common Lisp
CUDA, Apple silicon, and AMD
<https://github.com/cgore/chorus/>

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Introduction

Chorus is a library for GPU programming from Common Lisp. It targets NVIDIA GPUs through CUDA, Apple silicon, and AMD GPUs. It is a fork of `cl-cuda` by Masayuki Takagi [1].

The package `chorus` is the interface most programs use. `*backend*` and `list-devices` select a backend. On a Mac the default is Apple silicon when Metal is available. Where the CUDA driver loads, the default is CUDA. Native CUDA is the package `chorus/cuda`. Native Apple silicon is the package `chorus/apple-silicon`. It keeps a command queue, allocates shared buffers, compiles Metal Shading Language, and launches kernels. Its `defkernel` accepts `let`, `set`, `if`, `aref`, arithmetic, and the index variables. The chapters after this introduction describe the CUDA implementation: the kernel language, the driver API, and `with-cuda`.

The kernel language expresses kernel functions, kernel macros, and kernel symbol macros as S-expressions. On the CUDA backend, a kernel is compiled to CUDA C, then to PTX by `nvcc`, then loaded through the CUDA driver API. The kernel manager does this the first time the kernel is launched. A launch looks like an ordinary function call. It takes `:grid-dim` and `:block-dim`, and on the CUDA backend it runs inside a CUDA context.

Memory blocks pair a host allocation with a device allocation. `sync-memory-block` copies between them. The same package also exposes the host pointer and the device pointer separately.

Targets

- NVIDIA GPUs, through CUDA.
- Apple silicon.
- AMD GPUs.

The CUDA backend is verified on Windows and Linux with a current CUDA toolkit. An RTX 50-series GPU (Blackwell, compute capability 12.0) needs CUDA Toolkit 12.8 or later. CUDA 13.x is the toolkit this

backend is written for. With CUDA 13, `nvcc` can target Turing and newer (`sm_75` and above).

The repository is at <https://github.com/cgore/chorus/>.

Requirements for the CUDA backend

- An NVIDIA CUDA-capable GPU.
- The CUDA driver (`nvcuda.dll` on Windows, `libcuda.so.1` on Linux).
- The CUDA Toolkit, so that `nvcc` can compile kernels to PTX.
- On Windows, Visual Studio with the C++ workload, so that `nvcc` has a host compiler.

Kernel files are written to the operating system's temporary directory unless `*tmp-path*` is set. `nvcc` is found on `PATH`, through `CUDA_PATH` or `CUDA_HOME`, or in the usual toolkit locations.

Loading the library

Once the checkout is visible to ASDF:

```
(ql:quickload :chorus)
```

The test suite runs when `chorus-test` is loaded:

```
(asdf:test-system :chorus)
```

From a shell, in the checkout:

```
sbcl --load ~/quicklisp/setup.lisp --load test/run.lisp
```

A kernel

The following is the CUDA backend's form of `examples/vector-add.lisp`.

```
(defkernel vec-add-kernel (void ((a float*) (b float*) (c float*) (n int)))
  (let ((i (+ (* block-dim-x block-idx-x) thread-idx-x)))
    (if (< i n)
        (set (aref c i)
              (+ (aref a i) (aref b i)))))

(defun main ()
  (let* ((dev-id 0)
        (n 1024)
        (threads-per-block 256))
```

```

        (blocks-per-grid (/ n threads-per-block)))
(with-cuda (dev-id)
  (with-memory-blocks ((a 'float n)
                       (b 'float n)
                       (c 'float n))
    (random-init a n)
    (random-init b n)
    (sync-memory-block a :host-to-device)
    (sync-memory-block b :host-to-device)
    (vec-add-kernel a b c n
      :grid-dim (list blocks-per-grid 1 1)
      :block-dim (list threads-per-block 1 1))
    (sync-memory-block c :device-to-host)
    (verify-result a b c n))))

```

`aref` reads an array element. `set` writes one. `block-dim-x`, `block-idx-x`, and `thread-idx-x` are the kernel language's index variables. The CUDA backend maps them to the CUDA built-ins.

Chapter 1

The chorus Package

Load this package with `(ql:quickload :chorus)`. It reexports the generic backend interface from `chorus/backend`, the kernel language from `chorus/lang`, the CUDA API from `chorus/api`, and two driver variables from `chorus/driver-api`: `*show-messages*` and `*sdk-not-found*`.

`*backend*` and `list-devices` are the calls most programs make. `chorus/cuda` is the native CUDA package. `chorus/apple-silicon` is the native Apple silicon package.

`*sdk-not-found*` is true when the CUDA driver library does not load. Driver calls then signal `sdk-not-found-error`. Chorus also targets Apple silicon and AMD GPUs.

The packages below are the ones `chorus` is built from. Use them when you want a narrower set of symbols than the reexport.

Chapter 2

The chorus/api/context Package

This package initializes CUDA, selects a device, and keeps a context. `chorus/api` reexports it.

2.0.1 The `with-cuda` Macro

Syntax

```
(with-cuda (dev-id) &body body)
```

The macro calls `init-cuda`, binds `*cuda-device*` from `dev-id`, binds `*cuda-context*`, and runs `body`. If `*nvcc-options*` does not already name an architecture, the macro adds the `nvcc -arch=sm.XY` option for that device's compute capability. On the way out it unloads `*kernel-manager*` and destroys the context.

2.0.2 The `*cuda-device*` Variable

The device selected by `with-cuda`.

2.0.3 The `*cuda-context*` Variable

The context created by `with-cuda`.

2.0.4 The `*cuda-stream*` Variable

The stream passed to kernel launches. Its initial value is a null pointer.

2.0.5 The `init-cuda` Function

Calls `cu-init` with flags 0.

2.0.6 The `get-cuda-device` Function

Returns the device for an ordinal.

2.0.7 The `device-compute-capability` Function

Returns the compute capability of a device.

2.0.8 The `create-cuda-context` Function

Creates a context on a device.

2.0.9 The `destroy-cuda-context` Function

Destroys a context.

2.0.10 The `synchronize-context` Function

Waits until the work in the context has finished.

Chapter 3

The chorus/api/memory Package

This package allocates device memory, host memory, and memory blocks that pair the two. `chorus/api` reexports it.

3.1 Memory blocks

3.1.1 The `alloc-memory-block` Function

Syntax

```
(alloc-memory-block type size)
```

Allocates a block of *size* elements of *type*.

3.1.2 The `free-memory-block` Function

Frees both the device pointer and the host pointer of a block.

3.1.3 The `with-memory-block` Macro

Syntax

```
(with-memory-block (var type size) &body body)
```

Binds *var* to a new block and frees it on the way out.

3.1.4 The `with-memory-blocks` Macro

Syntax

```
(with-memory-blocks bindings &body body)
```

Each binding has the form `(var type size)`. The macro nests `with-memory-block`.

3.1.5 The sync-memory-block Function

Syntax

```
(sync-memory-block memory-block direction)
  direction is :host-to-device or :device-to-host.
```

3.1.6 The memory-block-aref Function

Syntax

```
(memory-block-aref memory-block index)
```

Reads the host side of the block. `setf` writes it. A copy to or from the device is a separate `sync-memory-block` call.

The package also exports `memory-block-p`, `memory-block-device-ptr`, `memory-block-host-ptr`, `memory-block-type`, and `memory-block-size`.

3.2 Separate host and device memory

`alloc-device-memory` takes a type and a count. `free-device-memory` frees that pointer. `with-device-memory` has the form `(with-device-memory (var type n) &body body)`.

`alloc-host-memory`, `free-host-memory`, and `host-memory-aref` are the host-side counterparts. `with-host-memory` has the same form as `with-device-memory`.

`memcpy-host-to-device` and `memcpy-device-to-host` copy a region. `device-total-bytes`, `device-total-kbytes`, `device-total-mbytes`, and `device-total-gbytes` report the memory on a device. `device-total-bytes` takes the device as its argument.

Chapter 4

The chorus/api/defkernel Package

This package defines kernels and launches them. `chorus/api` reexports it. It does not reexport `chorus/api/kernel-manager`. A kernel defined here is registered on `*kernel-manager*` and compiled the first time it is launched.

4.0.1 The `defkernel` Macro

Syntax

```
(defkernel name (return-type arguments) &body body)
```

Each argument has the form `(var type)`. The macro defines a Lisp function of those variables with keyword arguments `grid-dim` and `block-dim`. Both default to `(1 1 1)`. The launch uses `cu-launch-kernel` and the stream in `chorus/api/context: *cuda-stream*`.

4.0.2 The `defkernelmacro` Macro

Syntax

```
(defkernelmacro name arguments &body body)
```

Defines a macro in the kernel language.

4.0.3 The `defkernel-symbol-macro` Macro

Syntax

```
(defkernel-symbol-macro name expansion)
```

Defines a symbol macro in the kernel language.

4.0.4 The defglobal Macro

Syntax

```
(defglobal name expression &optional qualifiers)
```

Defines a kernel global. When *qualifiers* is omitted, the qualifier is `:device`.

4.0.5 The global-ref Function

Returns the device pointer of a global after ensuring it is loaded.

4.0.6 The expand-macro-1 Function

Expands one kernel-macro call.

4.0.7 The expand-macro Function

Expands kernel macros in a form.

4.1 The kernel manager

`chorus/api/kernel-manager` holds the definitions and the loaded CUDA module. `*kernel-manager*` is that object. `defkernel` calls `kernel-manager-define-function`. A `launch` calls `ensure-kernel-function-loaded`. `with-cuda` calls `kernel-manager-unload` when the context ends.

The package also exports `make-kernel-manager`, the `kernel-manager-*` readers, `kernel-manager-compile-module`, `kernel-manager-load-module`, `kernel-manager-load-function`, `kernel-manager-load-global`, and the other `ensure-kernel-*` operations.

Chapter 5

The chorus/api/macro Package

These are kernel macros, defined with `defkernelmacro`. `chorus/api` reexports them. A kernel that uses them should import or use this package. The test `chorus/test/api/smoke` shadows imports `let*`, `when`, and `unless` from here so that the kernel body sees the kernel versions.

5.0.1 The `let*` Macro

Binds sequentially. An empty binding list expands to `progn`.

Syntax

```
(let* bindings &body body)
```

5.0.2 The `when` Macro

Syntax

```
(when test &body body)  
Expands to (if test (progn body)).
```

5.0.3 The `unless` Macro

Syntax

```
(unless test &body body)  
Expands to (if (not test) (progn body)).
```


Chapter 6

The chorus/api/nvcc Package

This package finds `nvcc` and invokes it. `chorus/api` reexports `*tmp-path*`, `*nvcc-options*`, `*nvcc-binary*`, `find-nvcc`, `nvcc-available-p`, and `nvcc-arch-option`. `nvcc-compile`, `find-msvc-cl`, and `arch-option-p` stay in this package.

6.0.1 The `*tmp-path*` Variable

Directory for temporary `.cu` and `.ptx` files. `nil` means `uiop:temporary-directory`.

6.0.2 The `*nvcc-options*` Variable

Extra command-line options passed to `nvcc`.

6.0.3 The `*nvcc-binary*` Variable

Path to `nvcc`. `nil` or the string `nvcc` means auto-detect, using `PATH`, `CUDA_PATH`, and the usual toolkit locations on Windows and Unix.

6.0.4 The `nvcc-arch-option` Function

Syntax

`(nvcc-arch-option major minor)`

Returns an `-arch=sm_XY` option. Compute capability 12.0 becomes `sm_120`. `major` is a nonnegative integer. `minor` is an integer from 0 through 9.

6.0.5 The `nvcc-available-p` Function

True when an `nvcc` binary can be found.

6.0.6 The `find-nvcc` Function

Returns the `nvcc` pathname that auto-detect selected.

6.0.7 The `find-msvc-cl` Function

Returns the Windows host compiler `cl.exe` when one is found.

6.0.8 The `nvcc-compile` Function

Compiles a CUDA C string to PTX. The include path passed to `nvcc` is the `include/` directory of the `chorus` system.

6.0.9 The `arch-option-p` Function

True when an option list already contains an architecture selector.

Chapter 7

The chorus/api/timer Package

Timers are built from CUDA events. `chorus/api` reexports this package.

7.0.1 The `create-timer` Function

Syntax

```
(create-timer)
```

7.0.2 The `destroy-timer` Function

Destroys a timer.

7.0.3 The `start-timer` Function

Records the start event.

7.0.4 The `stop-timer` Function

Records the stop event.

7.0.5 The `synchronize-timer` Function

Waits for the timer's events.

7.0.6 The elapsed-time Function

Syntax

```
(elapsed-time timer)
```

7.0.7 The with-timer Macro

Syntax

```
(with-timer (var) &body body)
```

Binds *var* to a new timer and destroys it on the way out.

Chapter 8

The chorus/lang Package

This is the kernel language. `chorus` reexports it. Every generated `.cu` file includes `int.h`, `float.h`, `float3.h`, `float4.h`, `double.h`, `double3.h`, `double4.h`, and `curand.h` from the `include/` directory.

8.1 Index variables

`grid-dim-x`, `grid-dim-y`, `grid-dim-z`, `block-dim-x`, `block-dim-y`, `block-dim-z`, `block-idx-x`, `block-idx-y`, `block-idx-z`, `thread-idx-x`, `thread-idx-y`, and `thread-idx-z`.

8.2 Statements

`set` stores a value. `with-shared-memory` declares shared memory for the block.

8.3 Types

Scalar types: `void`, `bool`, `int`, `float`, `double`, and `curand-state-xorwow`.

Vector types, also available as Lisp structures: `float3`, `float4`, `double3`, and `double4`. Each has a constructor `make-float3` (and the same pattern for the others), slot readers, a predicate, an equality test, and a `with-float3` macro.

Pointer types: `bool*`, `int*`, `float*`, `double*`, `float3*`, `float4*`, `double3*`, `double4*`, and `curand-state-xorwow*`.

`cffi-type` maps a Chorus type to the CFFI type used at a launch.

8.4 Built-in operations

xor, shl, shr, rsqrt, __exp, __divide, atomic-add, pointer, syncthread, double-to-int-rn, dot, curand-init-xorwow, curand-uniform-float-xorwow, curand-uniform-double-xorwow, curand-normal-float-xorwow, and curand-normal-double-xorwow.

8.5 Compiler entry points

The compiler packages export one operation each: `compile-symbol`, `compile-type`, `type-of-expression`, `compile-expression`, `compile-statement`, and `compile-kernel`. `compile-kernel` is what the kernel manager calls.

Chapter 9

The chorus/driver-api Package

This package is the foreign binding to the CUDA driver API. Names follow the driver entry points with a `cu-` prefix and Lisp hyphens. When the driver library is absent, a call signals `sdk-not-found-error`.

9.1 Variables

9.1.1 The `*sdk-not-found*` Variable

True when the driver library did not load.

9.1.2 The `*show-messages*` Variable

Controls driver-API diagnostic messages. `chorus` reexports both variables.

9.2 Types

`cu-result`, `cu-device`, `cu-context`, `cu-module`, `cu-function`, `cu-stream`, `cu-event`, `cu-device-ptr`, and `size-t`.

9.3 Event flags

`cu-event-default`, `cu-event-blocking-sync`, `cu-event-disable-timing`, and `cu-event-interprocess`.

9.4 Functions

Device and context: `cu-init`, `cu-device-get`, `cu-device-get-count`, `cu-device-compute-capability`, `cu-device-get-attribute`, `cu-device-attribute-compute-capability`, `cu-device-attribute-compute-capability-minor`, `cu-device-get-name`, `cu-ctx-create`, `cu-ctx-destroy`, `cu-ctx-synchronize`, **and** `cu-device-total-mem`.

Memory: `cu-mem-alloc`, `cu-mem-free`, `cu-mem-host-register`, `cu-mem-host-unregister`, `cu-memcpy-host-to-device`, `cu-memcpy-host-to-device-async`, `cu-memcpy-device-to-host`, **and** `cu-memcpy-device-to-host-async`.

Modules and launch: `cu-module-load`, `cu-module-unload`, `cu-module-get-function`, `cu-module-get-global`, **and** `cu-launch-kernel`.

Events: `cu-event-create`, `cu-event-destroy`, `cu-event-elapsed-time`, `cu-event-record`, **and** `cu-event-synchronize`.

Streams: `cu-stream-create`, `cu-stream-destroy`, `cu-stream-query`, `cu-stream-synchronize`, **and** `cu-stream-wait-event`.

Chapter 10

The chorus/interop Package

This package is a separate system, `chorus-interop`. It depends on `chorus`, `cl-opengl`, `cl-glu`, and `cl-glut`. Load it with `(ql:quickload :chorus-interop)`.

It reexports `chorus/lang`, the interop API, and the driver variables `*show-messages*` and `*sdk-not-found*`.

10.1 Graphics driver entry points

`chorus/interop/driver-api` adds `cu-graphics-resource`, the `cu-graphics-register-flags-*` and `cu-graphics-map-resource-flags-*` enumerations, and these functions: `cu-gl-ctx-create`, `cu-graphics-gl-register-buffer`, `cu-graphics-map-resources`, `cu-graphics-resource-get-mapped-pointer`, `cu-graphics-resource-set-map-flags`, `cu-graphics-unmap-resources`, and `cu-graphics-unregister-resource`. It also reexports `chorus/driver-api`.

10.2 What the interop API replaces

`chorus/interop/api` reexports `chorus/api` except the `context`, `memory-block`, and `defkernel` symbols that the interop packages replace. The replacements are `create-cuda-context`, `with-cuda`, `defkernel`, and the memory-block operations, including `memory-block-vertex-buffer-object` and `memory-block-graphic-resource-ptr`.

The example `interop/examples/nbody.lisp` is the system `chorus-interop-examples`. Its package is `chorus/interop/examples/nbody`.

Bibliography

- [1] TAKAGI, M. cl-cuda. <https://github.com/takagi/cl-cuda>, 2012.